Legal Research and Evidence Assisstant-“JuriAid”

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Proposal Document

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

The Sri Lankan judicial system is hampered by outdated legal research methods, which take time. Retrieval of legal documents is often automated, but there are no advanced algorithms capable of understanding the full context of the documents and producing meaningful outputs. Solving this problem is challenging as most solutions do not account for the unique legal system of Sri Lanka.

This project proposes to develop a new system to find past court decisions similar to a new court decision based on their similarities using a knowledge graph. The system will use computational models to automatically classify legal document lexemes into basic logical roles (e.g., facts, issues) and obtain similarity through a weighted hybrid similarity metric based on these parts. The system will integrate citation links and their semantic significance into the knowledge graph to enable multi-hop reasoning to find indirect connections between cases. The system will be tailored to the Sri Lankan context, allowing for direct PDF uploads, English text processing, and interpretable outputs. The system seeks to improve the efficiency, accuracy, and accessibility of the Sri Lankan legal system.

Keywords: Legal Knowledge Graphs, Semantic Similarity, Rhetorical Role Identification, MultiHop Reasoning, Computational Linguistics, Sri Lankan Law, Case Similarity, Data-Driven Legal Tech.

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1.Introduction

In Sri Lanka, the legal system follows the doctrine of precedent, where earlier rulings guide how laws are interpreted and applied. lawyers rely on these decisions to support arguments and shape judgments in new cases.

However, finding and analyzing judgments is still slow and complex. Existing digital databases rely mainly on keyword searches, often missing cases that use different wording for similar reasoning.

To address this, the Past Case Retrieval Component is being developed within the Legal Research and Evidence Assistant system. It integrates NLP, Knowledge Graphs, Graph Neural Networks (GNNs), and Retrieval-Augmented Generation (RAG) to uncover precedents beyond direct citations or identical language.

A key feature is classifying sentences by rhetorical roles—Facts, Issues, Arguments, and Decisions—an approach shown by Sheetal et al. (2022) to improve case similarity over full-text or citation-based methods [1]. Knowledge Graphs further strengthen this by mapping legal entities (cases, statutes, judges, principles) and their relationships, enabling contextual understanding (Fensel et al., 2020) [2]. Such graphs have already enhanced retrieval, question answering, and recommendations in legal domains [3].

GNNs build on this by learning from graph structures to predict hidden links like case similarity or citations. Dhani et al. (2024) demonstrated that RGCNs are effective for Indian judgments when combined with expert-defined “law points” [3]. Similarly, this system will apply GNNs to Sri Lankan case law.

Finally, RAG grounds outputs in real precedents, ensuring that generated summaries and recommendations are both accurate and traceable.

Overall, the system aims to modernize legal research in Sri Lanka—making it faster, more precise, and transparent. By combining explainable AI with precedent-based reasoning, it will not only locate the most relevant cases but also justify their selection, ensuring both efficiency and trust.

2.Literature Review

Legal information retrieval has traditionally relied on keyword-based and citation-based search methods, which, while effective for exact matches, often fail to capture deeper semantic and contextual relationships between cases [4]. In the legal domain, especially for precedent-based systems such as in Sri Lanka, the ability to find cases with similar legal reasoning rather than identical terminology is critical.

One approach to improve retrieval effectiveness is rhetorical-role segmentation, where judgments are divided into functional sections such as *Facts*, *Issues*, *Arguments*, *Analysis*, and *Decision* [5]. Hierarchical sequence labeling models, such as SciBERT-HSLN, have been shown to accurately classify sentences into these roles, allowing retrieval systems to compare specific sections (e.g., *Facts-to-Facts*) rather than entire documents [6]. This method has consistently outperformed traditional document-levael similarity in legal contexts [7].

Research also highlights the benefits of role-aware similarity scoring, where textual similarity is calculated within comparable rhetorical segments. TF-IDF combined with cosine similarity has been found to yield strong performance, and giving higher weight to legally significant sections, such as *Facts* and *Analysis*, further improves retrieval accuracy [8].

Beyond text-based methods, Knowledge Graphs (KGs) have emerged as powerful tools for representing legal cases and their relationships [9]. In a KG, cases, statutes, legal concepts, and entities (e.g., parties, courts) are represented as interconnected nodes, with edges denoting relationships such as citations, legal principles, or case similarity. This graph-based representation allows for both efficient query execution and explainable results, as relationships between cases can be visualized [10].

Recent advances in Graph Neural Networks (GNNs), such as Relational Graph Convolutional Networks (R-GCN), have been applied to legal knowledge graphs for tasks like citation prediction and similarity-based ranking [11]. However, studies indicate that the performance of these models depends heavily on the availability of domain-specific features, such as legal concept annotations or local embeddings, which are currently lacking for Sri Lankan case law [12].

Retrieval-Augmented Generation (RAG) has also been explored in the legal domain to integrate semantic retrieval with natural language generation. In such systems, a vector database (e.g., FAISS, Pinecone) stores embeddings of legal documents, which are retrieved using models like Sentence-BERT or Legal-BERT [13]. The retrieved cases are then passed to a language model (e.g., GPT-4, LLaMA) to produce summaries or explanations grounded in the retrieved sources. While RAG primarily serves question answering or drafting, its retrieval component can be adapted directly for similar-case search [14].

Despite these advances, most existing work has focused on jurisdictions with publicly available datasets, such as the United States or India [15]. There is a lack of publicly accessible, structured, and annotated Sri Lankan legal corpora, especially those with rhetorical role segmentation.

Developing such resources would enable the adaptation of these state-of-the-art methods to the Sri Lankan context, improving both the accuracy and explainability of similar case retrieval.

3.Research Gaps

1. No Sri Lankan-Specific Legal Datasets Most existing legal AI systems rely on well-structured datasets from countries like India, the US, and Europe. Unfortunately, Sri Lanka doesn't yet have a publicly available, digitized, and properly organized collection of court judgments. This makes it difficult to train or test modern legal retrieval systems for local use.

2. Missing Role-Based Annotations in Sri Lankan Cases Breaking down judgments into structured sections—like Facts, Issues, Arguments, and Decisions—has been proven to improve case retrieval. However, no such annotated dataset exists for Sri Lankan case law. Without these labels, AI models can't compare cases based on their legal reasoning—only on superficial keyword matches.

3. Knowledge Graphs Haven't Been Applied to Sri Lankan Law Knowledge Graphs (KGs) are powerful tools for mapping connections between cases, laws, and legal concepts. While they've been used successfully in other countries, no one has yet built a Sri Lankan legal KG that combines case citations, semantic similarity, and role-based features.

4. Retrieval-Augmented Generation (RAG) Remains Untested for Sri Lankan Precedents RAG systems—which combine AI-powered search with natural language generation—are being used in places like the US and India for legal research. But no one has yet adapted this approach for Sri Lanka, especially not in a way that integrates role-based retrieval and knowledge graphs.

5. Most Systems Ignore PDF Processing—A Huge Practical Barrier Many legal AI tools assume clean, pre-extracted text. But in reality, lawyers and researchers work with PDFs—scanned documents, court filings, and unstructured case files. Without built-in PDF parsing, users have to manually extract text before running searches. This project solves that problem by allowing direct PDF uploads and automatic text processing for Sri Lankan cases..

4.Research Problem

Although legal case retrieval systems have shown some effectiveness in other countries, they perform poorly within the Sri Lankan legal context due to several key limitations. Most existing tools rely primarily on keyword matching, which fails to capture deeper semantic relationships in legal reasoning. Additionally, these systems overlook the rhetorical structure of judgments, treating them as unstructured text rather than differentiating between sections such as Facts, Issues, and Reasoning, which are crucial for accurate case similarity assessments.

Current retrieval approaches also tend to separate semantic similarity from citation networks, rather than integrating both, which limits their ability to uncover richer and more meaningful connections across precedents. Moreover, many AI-driven retrieval systems lack transparency, offering little explanation as to why a case was recommended—reducing trust and usability among legal practitioners. Finally, while knowledge graphs hold promise for capturing complex legal relationships, their rapid growth often leads to scalability and efficiency challenges, making them difficult to apply effectively in large collections of Sri Lankan case law.

5.Objectives

Main Objective

To develop a hybrid knowledge graph and vector database-driven retrieval system capable of input Sri Lankan court case PDFs and retrieving similar past cases with role-wise explainabilityB. Specific

Specific Objectives

1. Read Cases Directly from PDFs

Lawyers and judges work with PDFs — scanned documents, court filings, judgments. Right now, most AI tools can't handle them directly.

This system will let users upload a PDF of a Sri Lankan court case, and it will automatically pull out the text — cleanly and accurately — using reliable tools like PyMuPDF or pdfminer.

2. Break Down the Case Into Its Key Parts

Every legal judgment tells a story with structure:

What happened? → Facts

What's being disputed? → Issues

What did each side say? → Arguments

What was the final decision? → Decision

The system uses smart natural language processing (NLP) to automatically sort every sentence into these roles. This helps it understand the case like a lawyer would — not just as a block of text, but as a structured legal argument.

3. Build a Smart Legal Network

Instead of treating past cases like entries in a long list, this system organizes them like a web of connected legal knowledge think of it as a mind map for the law, built using Neo4j.

Each case is a node in the network.

4. Find Cases That Are Truly Similar — Even If They Use Different Words

It's not enough to match keywords. Two cases might talk about "unauthorized use of property" and "trespassing" different words, same idea.So the system uses smart AI models like Legal-BERT or Sentence-BERT to turn each part of a case into a meaning-based summary (embedding).These are stored in a vector database (like FAISS or Pinecone) so the system can quickly find cases that are semantically similar even if they don't share the same phrases.

5. Show Why a Case Was Recommended

AI shouldn't be a black box. If the system suggests a past case, you should know why.

So it doesn't just return a list — it explains:

6.Methodology

The development of this AI-powered legal question generation system will follow a systematic methodology, encompassing data collection, model development, system integration, and rigorous

testing. The overall system aims to provide a comprehensive solution for legal question automation.

Overall System

The whole system, as described by your context, can be conceptualized as an AI-powered legal assistant with a core component for question generation. Referring to the provided system diagram, the overall architecture of such a legal AI system would include:

- User Input: Lawyers provide initial case details, complaints, and replies [User's query, 135].
- Agent Framework Orchestrator: Manages task distribution and overall workflow.
- Court Decision Retrieval & Law Retrieval Engine: These components are responsible for fetching relevant statutes, precedents, and case law from a comprehensive legal knowledge graph (e.g., Neo4j) and embedding stores (using FATSS/SBERT and embeddings).
- Questionnaire Generator (Core Component): This module, powered by fine-tuned Legal-BERT with LoRA adaptation, dynamically generates context-aware questions and assesses legal positions.
- Legal Position Evaluation & Report Generator: Assesses the legal position based on input and generated questions, leading to output synthesis (e.g., draft questions, summaries).
- Long-Term Memory: Stores query history and aids in continuous learning.
- Final Output: Comprehensive legal analysis, guidance, and actionable recommendations (in your case, the generated legal questions).

OVERALL SYSTEM DIAGRAM

Figure 6.1 System Overview Diagram

Individual Component

This system helps lawyers quickly find past cases similar to a new one — not by keyword, but by meaning and legal structure. It's built for real use: upload a PDF, get smart, explainable results.

System work like,

1. Upload & Read PDFs

Start with a real court case PDF. The system pulls out the text (using PyMuPDF or pdfminer), cleans up junk like headers, and splits it into sentences.

2. Break It Down Like a Lawyer

Every sentence is tagged as Facts, Issues, Arguments, or Decision — using smart NLP models like Legal-BERT and SciBERT-HSLN. This helps the system understand the case the way a lawyer does.

3. Build a Legal Network (Knowledge Graph)

Using Neo4j, the system maps out cases, laws, parties, and how they're connected — like "this case cited that one" or "both deal with trespass." It's a web of legal knowledge.

4. Turn Text into Meaning (Vectors)

Key sections are converted into semantic vectors using Sentence-BERT and stored in a fast vector database (FAISS or Pinecone). That way, it can find cases that mean the same thing — even if they use different words.

5. Find Matches & Explain Why

When you search, the system finds similar cases using both meaning and citations. It shows why they match — like "Facts are 90% similar" — and highlights the relevant parts. A smart AI (like GPT-4) then summarizes the results in plain language.

6. Ready to Use

The backend runs on FastAPI/Flask, the frontend is built with React.js, and everything's packaged in Docker — so it works in the cloud (AWS/Azure) or on a local server.

INDIVIDUAL COMPONENT SYSTEM DIAGRAM

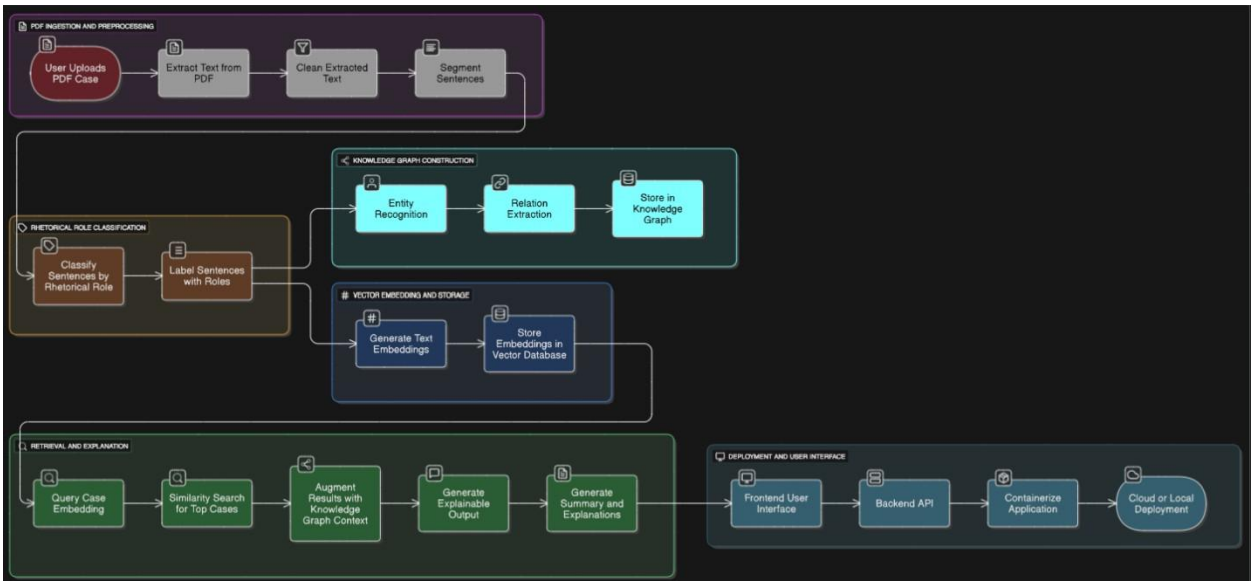

Figure 6.2 Component system diagram

7.Project Requirements

The system requirements for the Past Case Retrieval component are divided into two categories: Functional Requirements and Non-Functional Requirements.

A. Functional Requirements

The functional requirements define the core capabilities and operations that the system must perform.

1. PDF Case Upload – The system shall allow users to upload Sri Lankan court case documents in PDF format.
2. Text Extraction – The system shall extract text from PDF files using document parsing tools.
3. Text Preprocessing – The system shall remove headers, footers, and unwanted formatting, and segment text into sentences.
4. Rhetorical Role Classification – The system shall classify sentences into predefined roles such as Facts, Issues, Arguments, and Decision.
5. Knowledge Graph Construction – The system shall store extracted entities and relationships in a Neo4j-based knowledge graph.
6. Vector Embedding Generation – The system shall generate dense vector representations of legal text segments.
7. Vector Database Storage – The system shall store embeddings in a similarity search database (e.g., FAISS or Pinecone).
8. Case Similarity Retrieval – The system shall retrieve the top-N most similar past cases based on vector similarity.
9. Knowledge Graph Augmentation – The system shall enrich retrieval results with linked legal entities and citations.
10. Explainable Output Generation – The system shall present role-wise similarity scores, matched sections, and reasoning traces.
11. Summarization – The system shall generate natural language summaries of retrieved cases using an LLM within the RAG framework.
12. User Interface – The system shall provide a web-based interface for case upload, retrieval, and visualization.

B. Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints for the system.

1. Performance – The system shall return retrieval results within 5 seconds for typical query cases.
2. Scalability – The system shall handle up to 10,000 stored case documents without significant performance degradation.
3. Accuracy – The rhetorical role classification model shall achieve at least 85% F1-score on the test dataset.
4. Reliability – The system shall ensure 99% uptime during operational hours.
5. Security – Uploaded documents shall be stored securely, with access restricted to authorized users.
6. Maintainability – The system architecture shall support modular updates for NLP models and database components.
7. Usability – The web interface shall be intuitive and require no more than 10 minutes of user training.
8. Portability – The system shall be deployable on both cloud infrastructure (AWS/Azure) and local servers using Docker.
9. Compatibility – The system shall be compatible with major web browsers (Chrome, Firefox, Edge).
10. Data Privacy – The system shall comply with Sri Lankan legal data privacy and confidentiality standards.

8.Commercialization

Budget

The following amounts are estimated and may be subject to change:

Description	Cost	Occurring
Server Hosting	**LKR 15,000**	**Monthly**
Cloud Database	**LKR 20,000**	**Monthly**
Developer Tools/Software Subscriptions	**LKR 5,000**	**Monthly/Annual**
LLM API usage	**LKR 10,000**	**Monthly**

Table8.1budgettable

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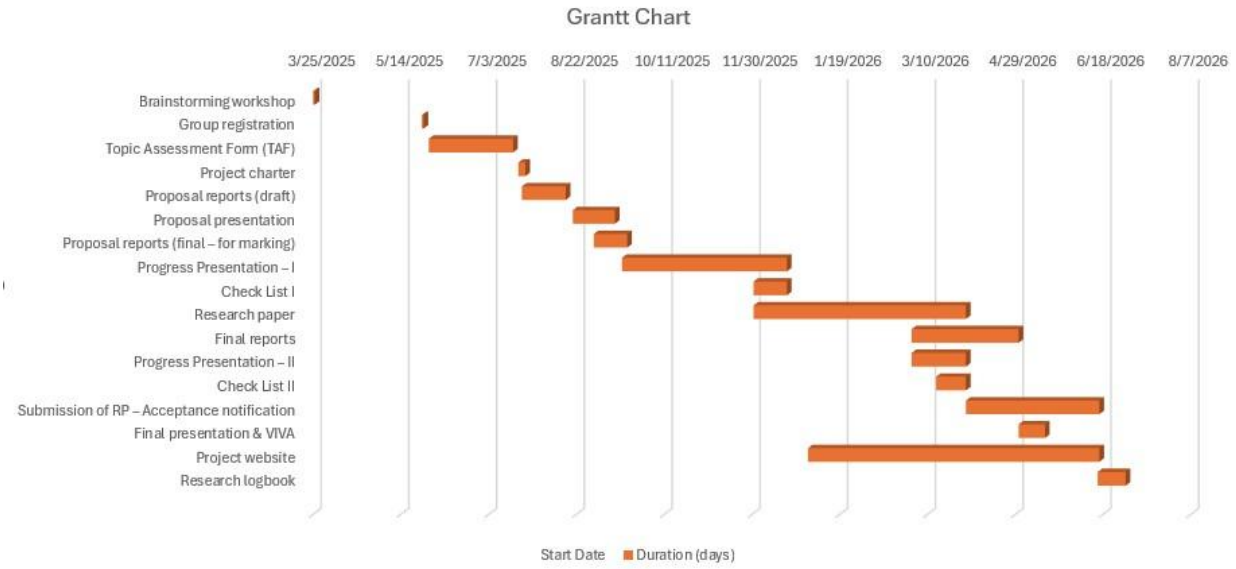
APPENDICES

Appendix 1. Gantt Chart